

Roll No. 2261514

End Semester Back Paper  
Examination 2022

Name of the Program: B.Tech.  
Name of the Course: Engineering  
Mathematics

Semester: II

Course Code: TMA-201

Time: 3 Hour's

Maximum Marks: 100

Note:

- All questions are compulsory.
- Answer any two sub questions among a, b, and c in each main question.
- Total marks in each main question are twenty.
- Each question carries 10 marks

|     |                                                                                                                                                                                                                                     |     |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Q1  | (10 X 2=20 marks)                                                                                                                                                                                                                   |     |
| (a) | Solve the differential equation<br>$(x + y - 10)dx + (x - y - 2)dy = 0$ .                                                                                                                                                           | CO1 |
| (b) | Solve using the Method of variation of parameters $(D^2 + 1)y = \sec x$                                                                                                                                                             |     |
| (c) | Obtain the general solution of $\frac{d^2y}{dx^2} - y = 3e^x - 2e^{2x}$                                                                                                                                                             |     |
| Q2  | (10 X 2=20 marks)                                                                                                                                                                                                                   |     |
| (a) | Find the Laplace transform of the function $\frac{\cos at - \cos bt}{t}$ .                                                                                                                                                          | CO2 |
| (b) | Obtain the inverse Laplace transform of the function $\cot^{-1}\left(\frac{s+3}{2}\right)$ .                                                                                                                                        |     |
| (c) | Applying Convolution theorem, solve the following initial value problem<br>$y'' + y = \sin 3t$ , given $y(0) = 0$ , $y'(0) = 0$ .                                                                                                   |     |
| Q3  | (10 X 2=20 marks)                                                                                                                                                                                                                   |     |
| (a) | Find the Fourier series for the function $f(x) = x$ in $(0, 2\pi)$ .                                                                                                                                                                | CO3 |
| (b) | Obtain the Fourier series for the function $f(x) = \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi \end{cases}$                                                                                                                |     |
| (c) | Obtain the Half range Fourier cosine series of the following function:<br>$f(x) =  x $ , $0 < x < \pi$ .                                                                                                                            |     |
| Q4  | (10 X 2=20 marks)                                                                                                                                                                                                                   |     |
| (a) | Solve the following partial differential equation:<br>$(D^2 - 2DD' + D'^2)z = \sin(x + 2y)$ .                                                                                                                                       | CO4 |
| (b) | Using the method of separation of variables, solve<br>$\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u$ , where $u(x, 0) = 6e^{-3x}$ .                                                                           |     |
| (c) | A tightly stretched string with fixed end points $x = 0$ and $x = l$ is initially in a position given by $y = y_0 \sin^3\left(\frac{\pi x}{l}\right)$ . If the string is released from this position, find displacement $y(x, t)$ . |     |

| Q5  | (10 X 2=20 marks)                                                                                  |  | CO5 |
|-----|----------------------------------------------------------------------------------------------------|--|-----|
| (a) | Show that $J_{1/2} = \sqrt{\frac{2}{\pi x}} \sin x$ and $J_{-1/2} = \sqrt{\frac{2}{\pi x}} \cos x$ |  |     |
| (b) | Prove the Rodrigue's formula $P_n(x) = \frac{1}{2^n n!} (x^2 - 1)^n$ .                             |  |     |
| (c) | Prove that $J_{-n}(x) = (-1)^n J_n(x)$ .                                                           |  |     |